

INTERNAL BRIEFING · CONFIDENTIAL

AI Voice Front Desk

From market research to a ready-to-ship plan — the complete journey, in four parts.

Part I · Pages 1–5 — How the market research was done

Part II · Pages 6–10 — The market: size, potential & the gap

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Prepared June 2026 · For internal review (founder + Shrikaanth) · Companion: Data Bible (21 tabs) + Build Bible. Every figure traces to committed, auditable source files.

How the Market Research Was Done

A rigorous, cross-validated demand × supply analysis — every dataset, keyword, and tool reviewed.

1.1 The Question & The Four-Lens Method

We did not start with an idea and then look for data to justify it. We started with a question — “*where is a real, winnable software gap we can build into?*” — and answered it with a four-lens analysis, deliberately cross-validated against bias.

Lens	What it answers	Data source
Demand	How many people want this? Is it growing? Seasonal?	Google Keyword Planner (4 years)
Winnability	Can we realistically rank for / win this?	Ubersuggest SEO difficulty
Commercial value	How much does the market pay to acquire here?	Keyword Planner cost-per-click
Incumbent weakness	Are the existing tools actually beatable?	164-tool teardown (two AI engines)

Only when a single territory scored well on *all four* lenses did we treat it as a real opportunity. That discipline is what separates an evidence-based decision from a hunch — and it is what this section documents.

Why two AI engines for the teardown? We ran the competitor analysis twice through different engines and treated a finding as high-confidence only where *both independently agreed*. This filters out hallucination and single-model bias. The most-corroborated finding — bad data quality plus pricing that punishes growth — became our strongest supply-side signal.

1.2 The Demand Data — Google Keyword Planner

We pulled four full years of search data (June 2022 – May 2026) across three markets.

For every keyword we captured: average monthly searches, three-month change, **year-over-year change**, and **the full 48-month series** — which let us compute seasonality and a true per-year trend, not just a single snapshot — plus competition and top-of-page CPC (low and high).

4 years

OF MONTHLY DATA

3

GEOGRAPHIES

48

DATA POINTS / KEYWORD

Geography	Keywords	Total avg searches / month
Worldwide	559	25,135,650
United States	517	6,013,630
Canada	302	910,840

How we read it

Search volume tells you demand *today*; the 4-year series tells you *direction*; CPC tells you *commercial intent* (advertisers only pay high CPCs where the lifetime value is high); and seasonality tells you *when* demand peaks. Together they form a far richer picture than a single volume number — and they exposed a flaw in our first read of the data, covered next.

1.3 The Keyword Universe & The 84% Correction

We built a structured set of **560 keywords** across 9 intent buckets — not a handful of guesses — so we could separate noise from genuine demand.

Bucket	Purpose	Bucket	Purpose
Category head terms	The software categories	All-in-one / consolidation	"Replace my many tools"
Problem / JTBD	What owners search before brands	Comparison / switching	"alternative to X" intent
ICP-qualified	"for small business / agency / teams"	Client-delivery	The post-sale ops layer
Agency-specific	Our natural lane	One-word head terms	crm, invoicing, scheduling...

The critical methodological catch: our first dataset looked enormous, but **~84% of the volume was branded search** — people typing the names of products that already won (notion, quickbooks, calendly). That measures fame, not unmet demand, and it is the trap most founders fall into. We re-pulled the research around *non-branded, problem-based* terms, then added **SEO difficulty** to all 560 — which is what finally surfaced the real, winnable opportunity.

This correction matters: it is the difference between "chase the famous category words and lose" and "find the winnable demand nobody is serving." The rest of Part I is built on the corrected, non-branded view.

1.4 The Tools We Reviewed — 164 Across 14 Categories

The supply side was a teardown of 164 SaaS tools — valuation, subscription pricing, architecture (AI-native vs legacy), go-to-market, and each tool's **biggest customer complaint**. Categories spanned CRM & sales automation, B2B data/prospecting, sales engagement, marketing automation, customer support, productivity/workspace, fintech, conversation intelligence, and all-in-one/agency.

Subscription pricing we recorded (sample)

Tool	Price	Tool	Price
Begin by Zoho	\$7–12 /user/mo	Apollo.io	\$49–119 /user/mo
Pipedrive	\$14–99 /user/mo	Close	\$49–139 /user/mo
Folk	\$18–39 /user/mo	Cognism	\$3,000–6,000+ /user/yr
Streak	\$15–129 /user/mo	GoHighLevel	CPC \$1,415 (highest)

Customer complaints (verbatim, then coded into themes)

Tool	Biggest complaint we recorded
Lusha	"Credits disappear fast on wrong or outdated rows"
Seamless.AI	"High volume of unverified or generic catch-all emails"
Close	"High telephony usage costs outside base monthly plan"
Folk	"Lacks advanced pipeline automation and conditional routing"

Valuations were recorded too (e.g., Pipedrive \$1.5B, Apollo.io \$1.6B) — confirming these are well-funded incumbents to out-flank, not out-spend.

1.5 SEO Difficulty & The Grey Space

Layering Ubersuggest SEO difficulty onto demand revealed which demand we could actually win — the single most important output of the research.

~70%

INCUMBENTS ARE LEGACY

16/16

BOTH ENGINES FLAG BAD DATA

24/79

COMPLAIN 'MISSING FEATURES'

Unwinnable walls	SEO	Winnable grey space	SEO	CPC
crm	90	project mgmt software for agencies	8	\$72
forms	93	crm with marketing automation	9	\$26
notion	77	social media reporting tool	10	\$106
booking	76	client retention software	15	\$126
leads	85	cloud based business phone system	9	\$76

The Opportunity Score — the synthesis

We combined all four lenses into a single ranking: *demand* × *low-SEO-difficulty* × *commercial value* × *incumbent weakness*. The winners clustered into one coherent territory at **SEO difficulty 8–35 with \$60–130 CPCs** — real demand, real money, and almost nobody ranking.

Part I conclusion: the data did not point to a crowded category — it pointed to the live, voice-first **customer-intake layer**: the highest CPCs, the single most open keyword (phone system, SEO 9), and the most-declining incumbents, all in one place. Part II sizes that opportunity.

The Market — Size, Potential & The Gap

How big it is, how big it can become, and exactly where the opening sits.

2.1 The Demand, Sized

25.1M

SEARCHES/MO WORLDWIDE

6.0M

US SEARCHES/MO

0.9M

CANADA SEARCHES/MO

The market is a power law: the top ~12 US keywords account for roughly 68% of volume, and those head terms are unwinnable walls. The opportunity lives in the winnable body and long tail — which is precisely where our Opportunity Score concentrated.

Why the headline number isn't the real number

25.1M worldwide searches sounds like the market — but ~84% is branded navigation. The **addressable, winnable demand** is the non-branded category and problem terms: still very large in aggregate, far less contested, and the basis for every sizing figure that follows.

Crucially, demand for the **job** our product does — answering, qualifying, booking — shows up across dozens of these non-branded terms (scheduling, appointments, phone system, client management, missed-call recovery), which is why the total addressable opportunity is a market, not a niche.

2.2 Where Attention Is Moving (4-Year Momentum)

The 48-month series shows the direction of travel — money and attention are flowing toward all-in-one, AI-native, and modern tools, and away from legacy email and support software.

Rising	4-yr trend	Signal
brevo	+6,320%	all-in-one marketing surging
gohighlevel	+573%	agency all-in-one (highest CPC)
apollo.io	+346%	AI-native data + outreach
close	+116%	calling-centric CRM

Declining	4-yr trend	Signal
convertkit	−70%	standalone email collapsing
freshdesk	−45%	legacy support fading
mailchimp	−33%	post-acquisition decline
zendesk	−24%	legacy support fading

The signal: the hottest, highest-CPC money is in agency all-in-one and customer-intake; the fastest-declining incumbents are legacy support and email tools — the exact zone our product attacks with an AI-native approach.

2.3 Market Size — Beachhead to Platform (US + Canada)

Layer	Size (estimate)	Meaning
Beachhead — physiotherapy	~40,000 clinics · ~\$190–220M	Where we start (we own 7)
Adjacent healthcare	~450,000+ locations · ~\$2B+	Dental, chiro, vet, med-spa
All appointment / phone-driven local business	2–3M+ businesses · \$10–15B+ TAM	The platform (3–4× globally)

Physiotherapy alone is a ~\$20–40M revenue niche — a great business, not a unicorn. The **multi-vertical local AI front desk is the \$10–15B+ opportunity**, and it is what makes a billion-dollar outcome credible.

The math that matters

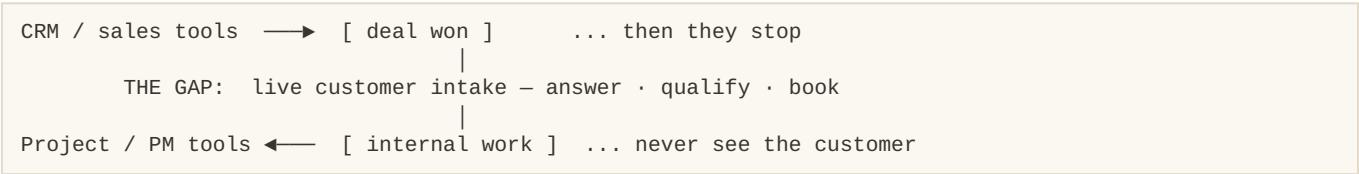
\$100M of annual recurring revenue is roughly **20,000 locations** at our pricing — a tiny slice of a 2–3M-business market. The constraint on reaching it is execution and distribution, not market size. We start where our distribution is unfair (our own clinics and networks) and expand vertical by vertical.

Why we start narrow: a single vertical caps at tens of millions; the platform is the billion-dollar story. Physiotherapy proves the product; construction (our GC clients) proves it is repeatable; the platform is the sum.

2.4 The Gap Analysis — Why This Exact Spot

Crossing demand with supply, three structural facts created the opening.

- 1. **~70% of incumbents are legacy** — AI bolted on after the fact, not built in. An AI-native entrant has structural room.
- 2. **The loudest, most-corroborated complaints** are bad data, punishing pricing, tool sprawl, and complexity — every one of them beatable by design.
- 3. **An unowned middle:** traditional CRMs stop at "deal won"; project tools are internal-only and don't understand customers. The **live customer-intake layer** — answering the phone, qualifying, booking — is under-served by modern, AI-native software.



All four research lenses converged here: highest CPCs, the single most open keyword (phone system, SEO 9), declining incumbents, and a clear "why now." That convergence — not a guess — is why we chose the AI Voice Front Desk.

2.5 Why Now, and Who Proves the Path

Why now

Real-time voice AI only crossed the quality threshold in **2025** — natural turn-taking, instant interruption handling, and sub-second latency finally made it possible to replace a human on the phone. The enabling infrastructure is now rentable, so a small team can ship in weeks while building the defensible moat on top.

The comparables — the path is already proven

Company	What they prove	Outcome
Podium	Local-SMB messaging / comms	~\$3B valuation
Weave	Healthcare comms (dental / medical)	Public company
ServiceTitan	Home-services vertical SaaS	~\$9B (public)

Local-SMB vertical software reliably produces multi-billion outcomes. AI is the wedge to take share from the legacy versions — and we attack a layer (the phone / front desk) none of them fully own with AI yet.

The burning problem that funds it all: local businesses lose real revenue every day on **missed calls** — each one a lost \$200–\$15,000 customer. We sell *recovered revenue*, provable in week one. That is the easiest enterprise sale there is.

"We didn't pick this market — the data did. Then we confirmed someone had already walked the path to a billion dollars. They had."

The Financials — Why It Makes Sense

Unit economics, pricing, projections, cost to build, and why the economics compound.

3.1 Unit Economics — The Engine

Delivery cost is usage-based and low; pricing is healthy SaaS. The gap between them is a 70–80% gross margin.

Cost component (per minute)	Tool	~Cost
Telephony	Twilio / Telnyx	\$0.005–0.015
Speech-to-text	Deepgram / AssemblyAI	~\$0.008
The brain (LLM)	Claude (Haiku mostly)	\$0.01–0.04
Text-to-speech	Cartesia / ElevenLabs	\$0.02–0.04
Orchestration	Retell → self-host later	\$0–0.07
Total delivery cost		~\$0.05–0.17 / minute

70–80%

GROSS MARGIN

\$75–225

COGS / LOCATION / MO

falling

COGS AS MODELS CHEAPEN

At a typical 500–1,500 minutes per location per month, delivery costs ~\$75–225 against \$300–800 in revenue. As foundation-model prices fall, **our costs drop while pricing holds** — so margins expand over time rather than compress.

3.2 Pricing Models — Multiple Ways to Capture Value

Model	Structure	Why
Tiered subscription	\$199 / \$399 / \$799 per location/mo	Core, predictable revenue
Setup fee	\$300–500 one-time	Instant cash; near-pure margin
Per-booked-appointment	\$15–40 per qualified booking	Pay-for-results; converts skeptics
White-label / agency	\$1,000–2,000/mo + per-minute	Second revenue stream + distribution
Vertical "Brain" packs	+\$100–300/mo add-on	Monetize the moat directly

Sample per-location monthly P&L

Line	Amount
Subscription (Growth tier)	\$399
Delivery cost (~1,500 min)	–\$150
Gross profit / location / mo	~\$249 (~62%+; higher at lower usage)
One-time setup (amortized)	+\$300–500 up front

The per-appointment option is the conversion weapon: at $\$25 \times 60$ booked/mo = \$1,500 against ~\$225 cost $\approx$ 85% margin — and the customer happily pays because they only pay for results.

3.3 Revenue Model & Illustrative Projection

Revenue scales with locations live. It is recurring, sticky (integrations create lock-in), and expands per account over time.

Stage	Locations	~ARR
Year 1 — physio + agency pilot	50–150	\$0.25–0.7M
Year 2 — + construction (our GCs)	400–800	\$2–4M
Year 3 — adjacent healthcare	2,000–4,000	\$10–20M
Year 5 — multi-vertical platform	15,000–25,000	\$75–125M ARR

What drives the curve

- **Land:** own businesses + owner-direct networks → cheap first customers.
- **Expand:** more locations per logo (multi-site groups), upsells (reporting, vertical packs), and white-label resellers.
- **Retain:** high switching cost once we run the front desk → low churn → revenue compounds.

Illustrative planning model, not a forecast. \$100M ARR ≈ ~20,000 locations — a small fraction of a 2–3M-business market.

3.4 Cost to Build & Run — Capital-Light

\$5-15k

TO FIRST PAYING CUSTOMER

\$500-1.6k/mo

TO RUN OUR 7 BUSINESSES

4-6 wks

TO MVP

Item	Cost
Infra & tools during build / pilot	~\$300–800 / month
Running the AI across our 7 own businesses	~\$500–1,600 / month (usage)
Legal one-time (consent, ToS, data-rights, vendor BAAs)	~\$1,500–5,000
People	Shrikaanth + AI executor agents (no new hires to start)

This is one of the cheapest starts possible for a venture-scale software company: the hard real-time engineering is rented (and improving for free), the first customers are our own businesses, and the workforce is largely AI agents supervised by one person.

Funding strategy: bootstrap on our businesses' cash to product-market fit, then raise from a position of proof — not a position of hope. The small capital requirement keeps ownership and optionality with us.

3.5 Why the Economics Compound

- **Low CAC:** our own businesses + owner-direct networks + the most open SEO keyword (difficulty 9) = inexpensive acquisition where competitors pay \$76 CPCs.
- **High retention / NRR:** once we answer the phones and write into their systems, switching cost is high and usage grows per account — the basis for net revenue retention above 100%.
- **Expanding margin:** model costs fall over time; the self-improving loop reduces failures (and therefore human QA cost) as we scale.
- **Operating leverage:** AI agents do the work; one human supervises. Revenue can grow far faster than headcount.

The financial thesis in one line: a capital-light start, 70–80% gross margins, low CAC via unfair distribution, and a sticky, expanding revenue base — pointed at a \$10–15B+ market with a proven path to multi-billion outcomes.

"The economics aren't a hope — they're a low-cost, high-margin engine attached to a market someone already proved is worth billions."

Architecture & Technical — How It Ships

The product, the system, how it self-improves, the lean operating model, and the build plan.

4.1 The Product — One Engine, Two Modes

Model 1 — Inbound Receptionist

Answers calls to the business 24/7, answers from its own knowledge base, qualifies the caller, and books into the calendar — with SMS confirmation and human handoff when unsure.

Model 2 — Speed-to-Lead Caller

When a lead submits a paid-ad form (with consent), the AI calls back within 60 seconds, qualifies, and books — capturing the ~390% conversion advantage of the 5-minute rule that no human team can match around the clock.

Both modes are the **same engine** — brain + knowledge base + booking adapter. Model 2 is simply Model 1 plus a lead-webhook trigger. We build once and point it at inbound calls, lead callbacks, or both.

4.2 The Architecture — Live vs Offline

```
LIVE CALL (latency budget < 800 ms – must feel human)
  Telephony – STT(entity-tuned) – Brain(Claude+RAG) – TTS – Telephony
    + barge-in (stop instantly on interrupt)
    + repeat-back confirm + DTMF keypad for phone digits
    + tentative booking to calendar
```

4.2 (cont.) — The Offline Layer & The Modules

OFFLINE (seconds after hangup – no latency pressure)

LLM polish + validate (normalize name, phone, email)

cross-check booking vs transcript → write to CRM

SMS confirm if any field low-confidence

score the call → feed the self-improvement loop

This split is the key design decision: the caller gets a fast, natural call; the heavy, accurate LLM validation runs *after*, so accuracy never costs latency.

Module	Job	Tool (Phase 1)
Telephony	Carry the call	Twilio / Telnyx
STT (ears)	Speech → text, entities + accent	AssemblyAI / Deepgram
Brain	Reason, qualify, decide (reads KB)	Claude (Haiku → Sonnet)
TTS (mouth)	Natural voice, stop on interrupt	Cartesia / ElevenLabs
Orchestration	Real-time wiring + barge-in	Retell → LiveKit (Phase 2)
Booking + SMS	Write to calendar, confirm	Google Cal / Jane / Jobber + Twilio
Capture + Eval	Log every call, score it	Supabase + Coval

Every module sits behind a provider-abstraction layer, so Phase 2 (owning barge-in + a multi-vendor router + a fine-tuned accent model on LiveKit/Pipecat) is a swap, not a rewrite.

4.3 How It Self-Improves (Auto-Correct)

The model gets smarter every night through a closed, eval-gated loop — with a human approving changes.

```
1 CAPTURE    every call → transcript + audio + outcome + confidences
2 AUTO-EVAL  LLM-as-judge scores it: booked? mis-heard a number?
              interrupted rudely? understood the accent?
3 MINE       cluster the failures into patterns
4 PROPOSE    fast (no training): update knowledge base, add mis-heard
              words to STT vocabulary, add a winning call as an example
              slow (periodic): fine-tune an accent model on our own calls
5 GATE       replay vs a saved test-set → ship only if better & safe
6 APPROVE    Shrikaanth approves → live.    Loop repeats nightly.
```

This is also the moat: the loop runs **per vertical**. Every physio call (anonymized) improves the "Physio Brain"; each new client makes the product smarter for the next — a data network effect no fresh competitor can match, and no foundation-model company will build per-vertical.

Honest boundary: fully autonomous self-editing is unsafe today, so the design is *auto-detect* → *auto-propose* → *eval-gate* → *human-approve*. The eval gate is the seatbelt that makes auto-improvement safe; as trust grows, more changes ship automatically.

4.4 How One Person Runs Everything

One human (Shrikaanth) orchestrates a team of AI executor agents and acts as reviewer/approver. The agents do the repetitive work.

Executor agent	Role
Dev Agent	Builds the app, integrations, webhooks (via Claude Code)
Onboarding Agent	Fetches client domain + Google Business Profile + uploaded docs → drafts the config → client confirms → live in under a day
QA / Eval Agent	Scores every call; surfaces only the ones needing human eyes
Improvement Agent	Proposes knowledge / vocabulary / prompt fixes for approval
Follow-up · Sales-Research · Monitoring	Post-call SMS drafts · prospect outreach · uptime/error alerts

The first additional human hire comes only at ~30–40 customers. The structural bet: **AI agents are the workforce; the human is the orchestrator.** This is what makes a billion-dollar-scale ambition feasible for a team that starts as one.

Onboarding in under a day: a client enters their domain + Google profile + documents; the Onboarding Agent auto-drafts an 80%-complete configuration from the vertical Brain template; the client confirms; three automated test calls run; Shrikaanth approves; it goes live after-hours.

4.5 Build Roadmap, Compliance & Ship Readiness

Phase	What ships	When
0 — Foundation	Stack + abstraction layer + capture schema + vendor BAAs + consent/disclosure	Wk 0–1
1 — Agency pilot	Speed-to-lead on PPC Guru leads → measure conversion lift	Wk 1–2
2 — Physio inbound	Physio Brain + Jane integration + after-hours, one clinic	Wk 2–4
3 — Self-improving + onboarding	Offline loop + executor agents + <1-day onboarding	Wk 3–6
4 — Go live + first external	Full receptionist + billing + first paying customer	Mo 2
5 — Scale	Construction + white-label + self-build migration	Mo 3+

Compliance by design

Consent, AI disclosure, call-recording, voiceprint/biometric, and HIPAA/BAA are handled in Phase 0. The AI qualifies and books but **never gives regulated advice**; an anti-bias review precedes any finance vertical. The full US + Canada regulatory register lives in the Build Bible.

Our USP

Vertical-deep (not generic) · actually understands the caller (accent-adapted) · books correctly (validated data capture) · self-improving + compliant by design · run by one person plus AI agents.

"Research-validated, financially sound, technically ready — built to ship in weeks, get smarter every night, and run by a team of one plus its agents."

End of briefing · Full detail in the committed Data Bible (21 tabs) and Build Bible.